MATLAB for Geoscience Workflows

Hydrology, Mapping, and Beyond

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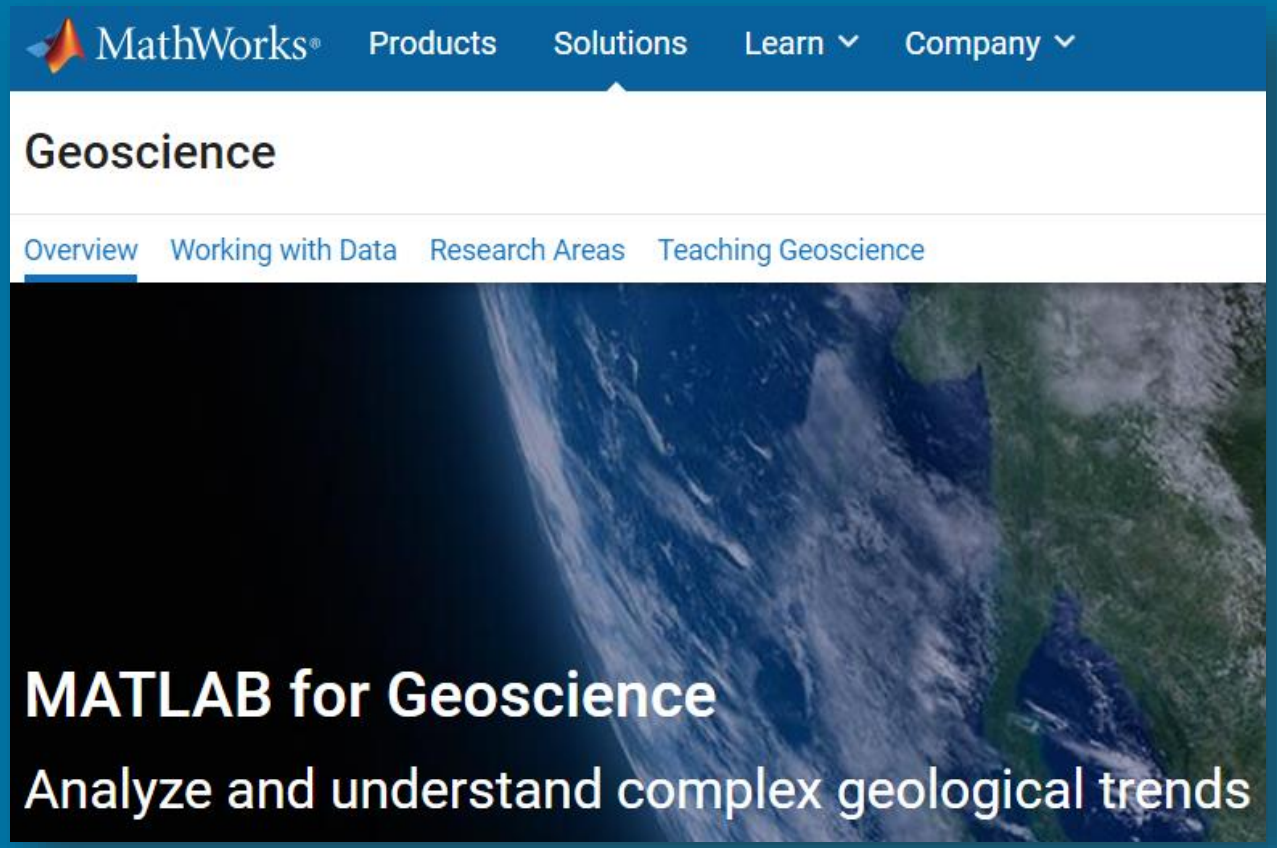
Check out the MATLAB for Geoscience webpage

- Geoscience Overview
- Working with Geoscience Data
- Research Areas
  - Seismology and Geophysics
  - Ocean and Climate
  - Hydrology and Water Science
  - Agriculture
- Teaching Geoscience



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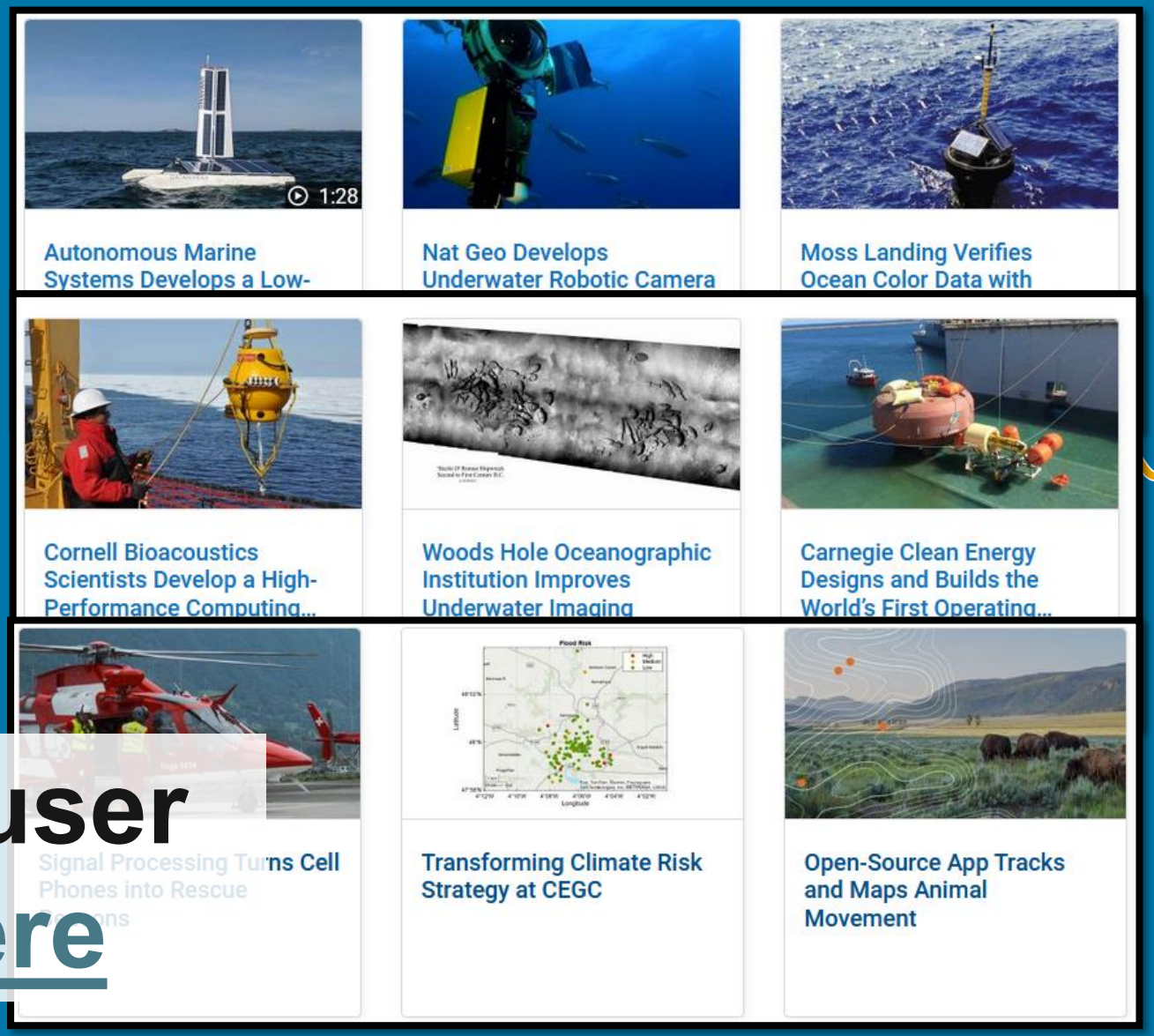

Find more user stories here

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Quick MATLAB Review

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Create interactive, human readable notebooks in the Live Editor

Make scripts readable

- Richly formatted text
- Interleave code & output

Add interactivity

- Hyperlinks and table of contents
- Live Controls to adjust parameters

Share with non-MATLAB users

- Export as report (pdf, docx, etc.)
- Convert to Jupyter notebook
- Share through GitHub and MATLAB Online

SunriseSunset.mlx

Estimating Sunrise and Sunset

We can estimate sunrise and sunset times if we know the latitude, longitude, and UTC offset. The solar declination (δ) is the angle of the sun relative to the earth's equatorial plane. On any given day of the year (d), using the latitude (ϕ), the sun's declination (δ) and the solar time correction (SC) we can calculate sunrise and sunset times.

Solar Declination	Sunrise	Sunset
$\delta = \sin^{-1}\left[\sin(23.45)\sin\left(\frac{360}{365}(d-81)\right)\right]$	$12 - \frac{\cos^{-1}(-\tan\phi\tan\delta)}{15^\circ} - \frac{SC}{60}$	$12 + \frac{\cos^{-1}(-\tan\phi\tan\delta)}{15^\circ} - \frac{SC}{60}$

Table of Contents

Estimating the Sunrise and Sunset Times
Plot Yearly Results

Estimating the Sunrise and Sunset Times

Set the latitude, longitude, and UT offset.

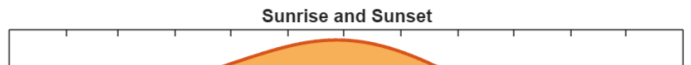
```
lat  =  42 ;
lon =  -71 ;
UTCoff = -5 ;
```

Estimate the sunrise and sunset times. We use the custom `equationOfTime` function to calculate the solar time corection (SC).

```
day = 1:365;
timeCorr = equationOfTime(day)
```

```
timeCorr = 1×365
    -3.7052   -4.1497   -4.5894   -5.0239   -5.4528   -5.8756   -6.2921   -6.7018 ...
```

```
plotSunriseSunset(lon, UTCoff, timeCorr, day, lat)
```



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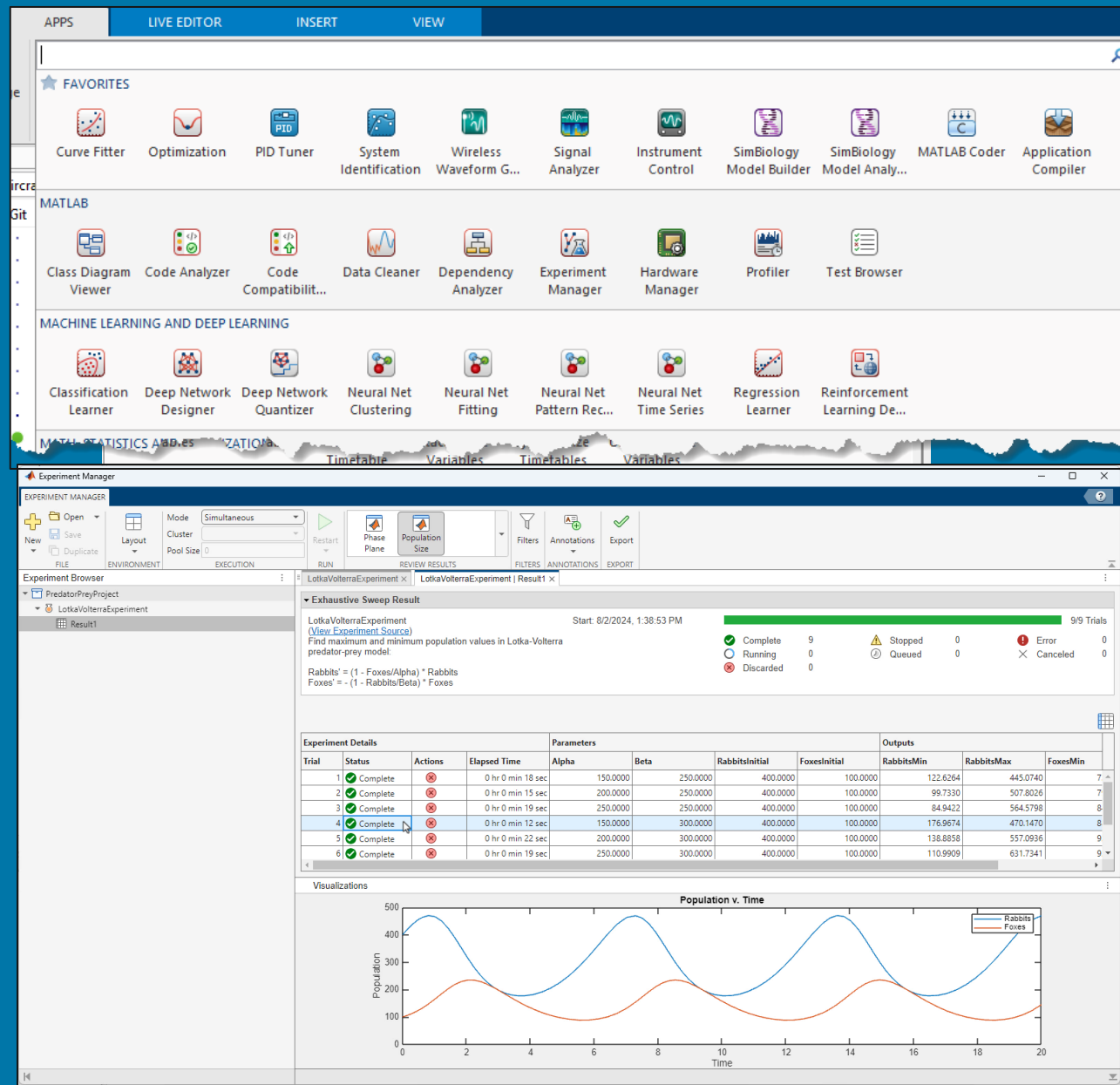
Interactively solve tasks and workflows

- Live Tasks solve steps within the Live Editor
- Apps solve larger, multi-step workflows
- Generate the code for your solution
- 40+ Live Tasks and 120+ Apps
- Build and share custom Tasks and Apps



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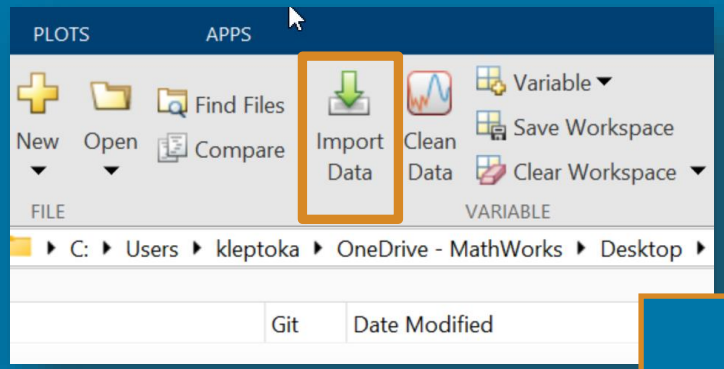
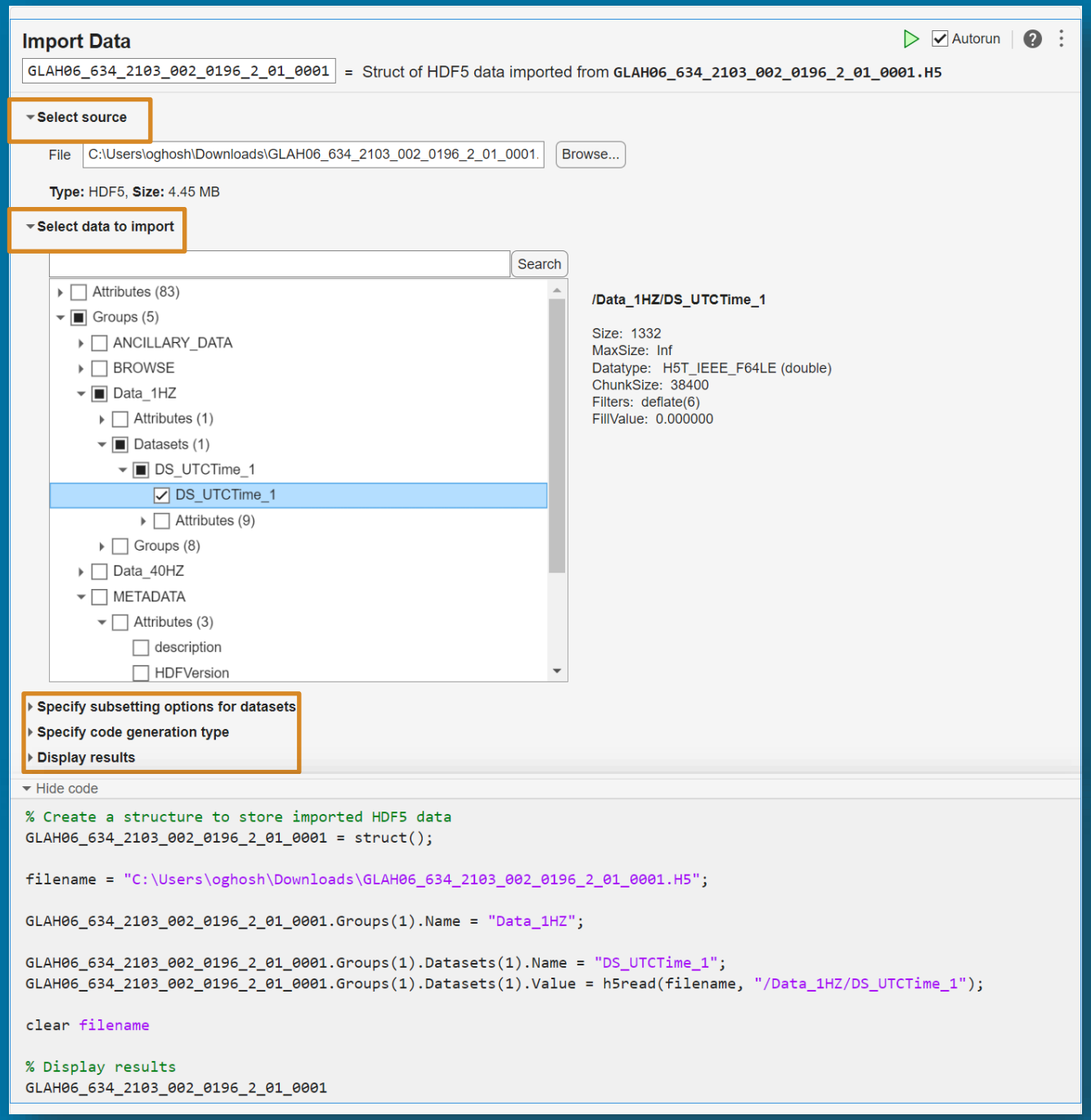
Data Handling in MATLAB: Import Data Live Task

supports:

- ***NetCDF (R2023b)***
- ***HDF5 (R2024a)***
- ***GRIB (R2024b)***
- ***And more***



Code Generated



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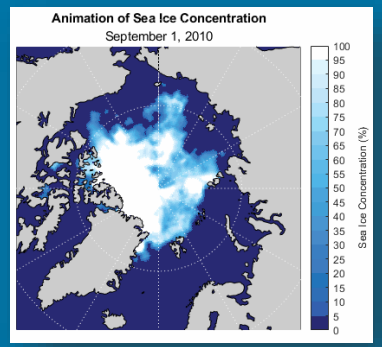
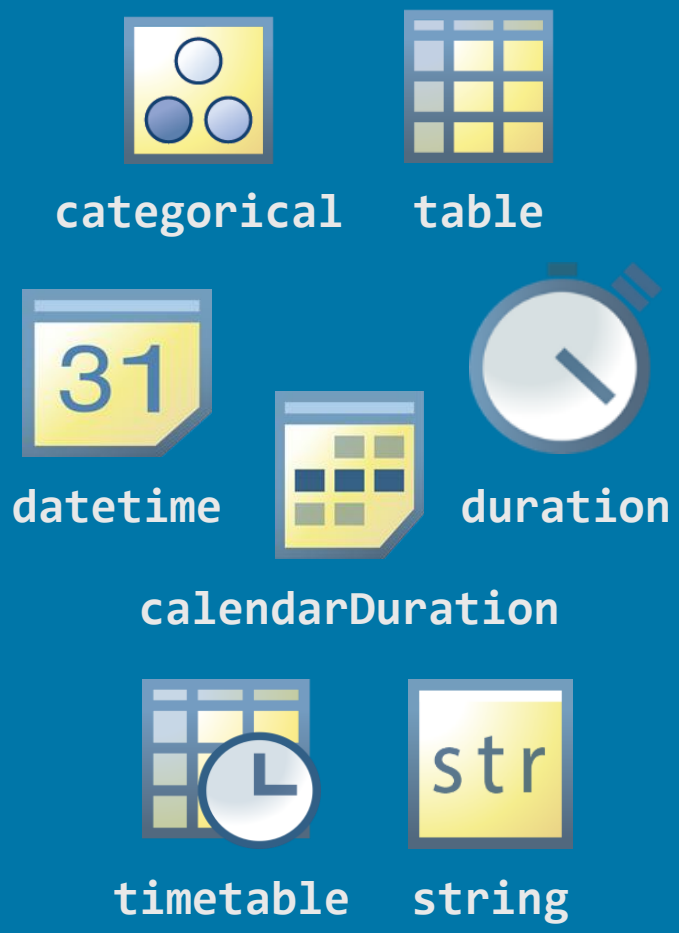
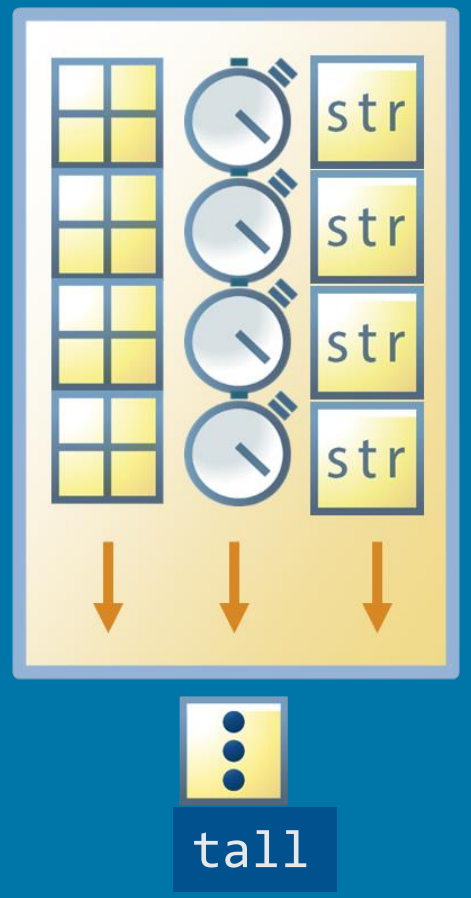
Supported data formats and types in MATLAB

- ***Raster file formats***: GeoTIFF, USGS DEM, DEM, DTED, Arc ASCII Grid, GTOPO30, ETOPO, and worldfile
- ***Vector file formats***: ESRI shapefiles, KML, GPX, VMAP0, GSHHS, and .gpkg
- ***Image file formats***: TIFF, JPEG, PNG, JPEG2000, LandSat, NITF and HDR
- ***Scientific data***: NetCDF, HDF5, GRIB, CDF, FITS, and multiband files (BIP, BIL, BSQ)
- Community Tools to work SAC and SEG-Y







Data Formats >>

Data Format Functions >>

Tall Arrays >>

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MATLAB Copilot

AI Assistant Optimized for MATLAB

New 2025!

```
%% Explore data
% A MAT-file is a binary MATLAB file that stores workspace variables.
% Use whos to list all variables in the workspace, including their type and
% size
whos

%% Process data
% Compute elapsed durations between entryTimestamps and exitTimestamps
durations = exitTimestamps-entryTimestamps;
% Compute the number of durations
numberOfDurations = numel(durations);
% Compute the average duration
averageDuration = 0;
for ii = 1:numberOfDurations
    averageDuration = averageDuration + durations(ii);
end
averageDuration = averageDuration/numberOfDurations;

%% Visualize data
% Plot all durations
figure, plot(durations)
% Add a horizontal line with the average duration over the current plot
hold on
plot([1,numberOfDurations],[averageDuration, averageDuration])
hold off
```

```
>> evaluateProcessedItems
  Name                  Size            Bytes  Class       Attributes

  entryTimestamps      100x1             1600  datetime
  exitTimestamps       100x1             1600  datetime

>> Press Ctrl + Shift + P to generate code with Copilot
```

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Find Hydrology Tools, Resources

- Geoscience Overview
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  - Ocean and Climate
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  - Agriculture
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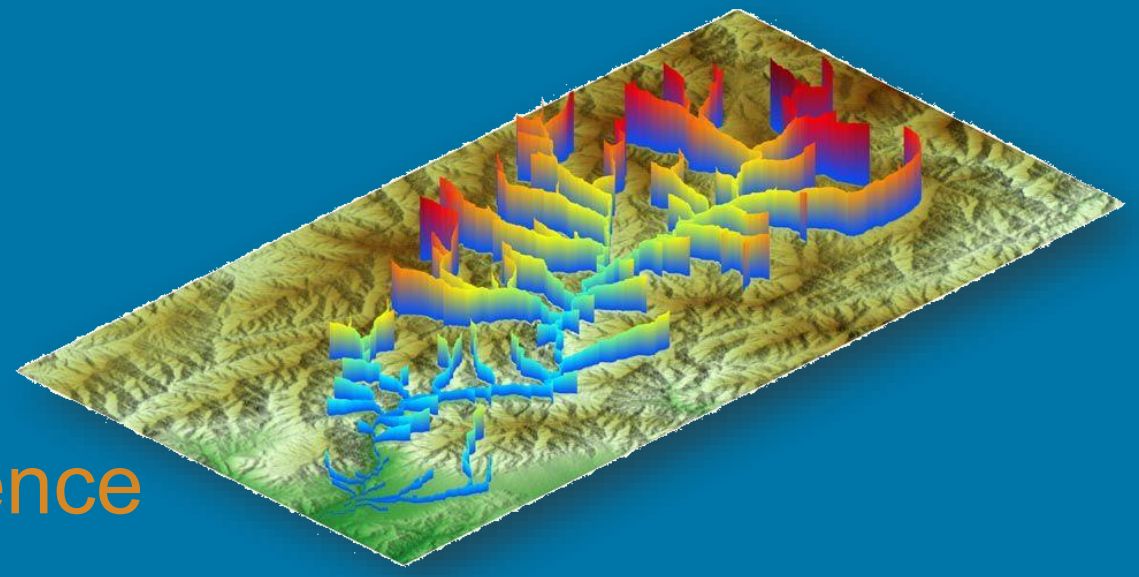

Topotoolbox: open-source toolbox for analyzing topographic data

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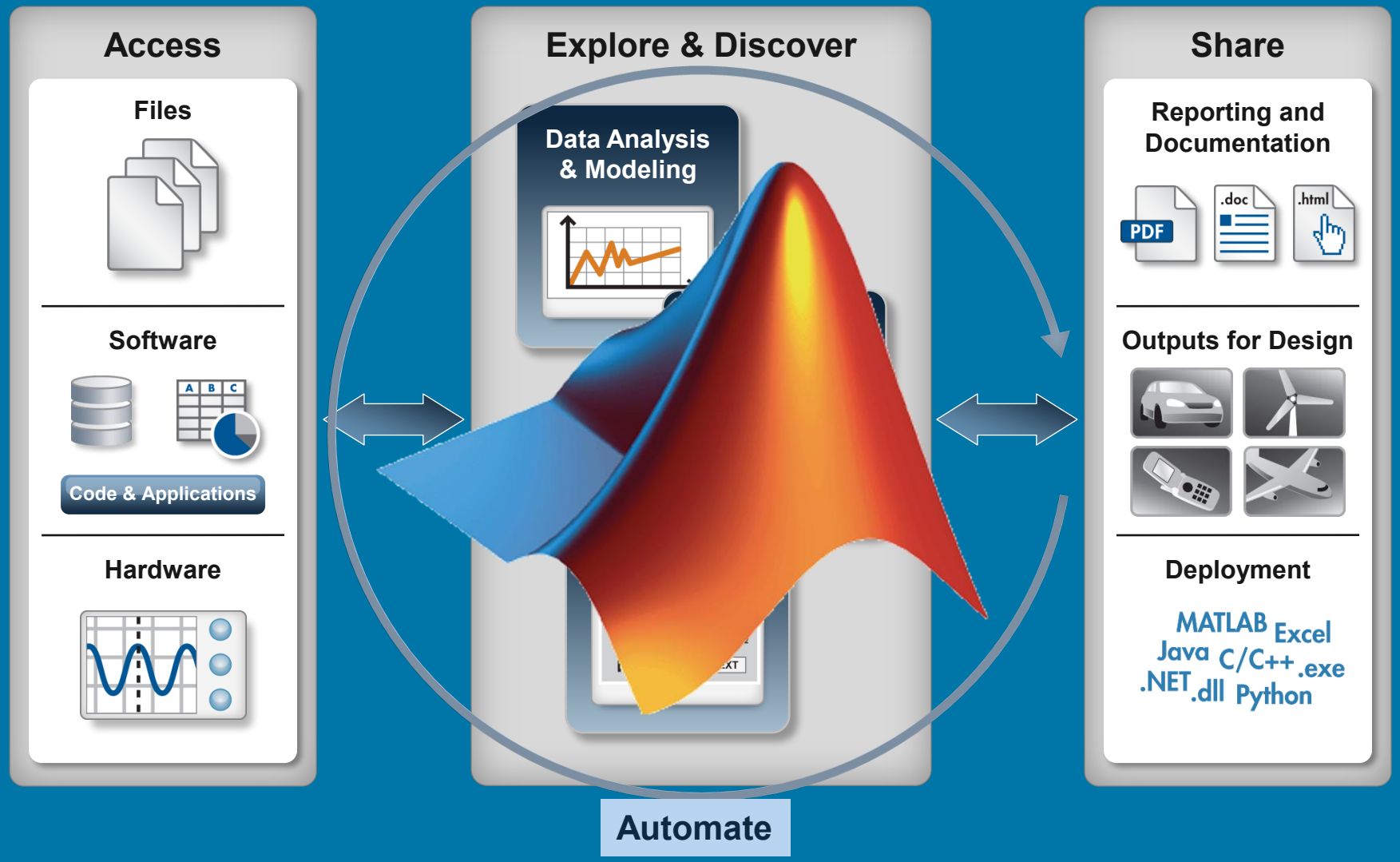
Data Analysis Workflow



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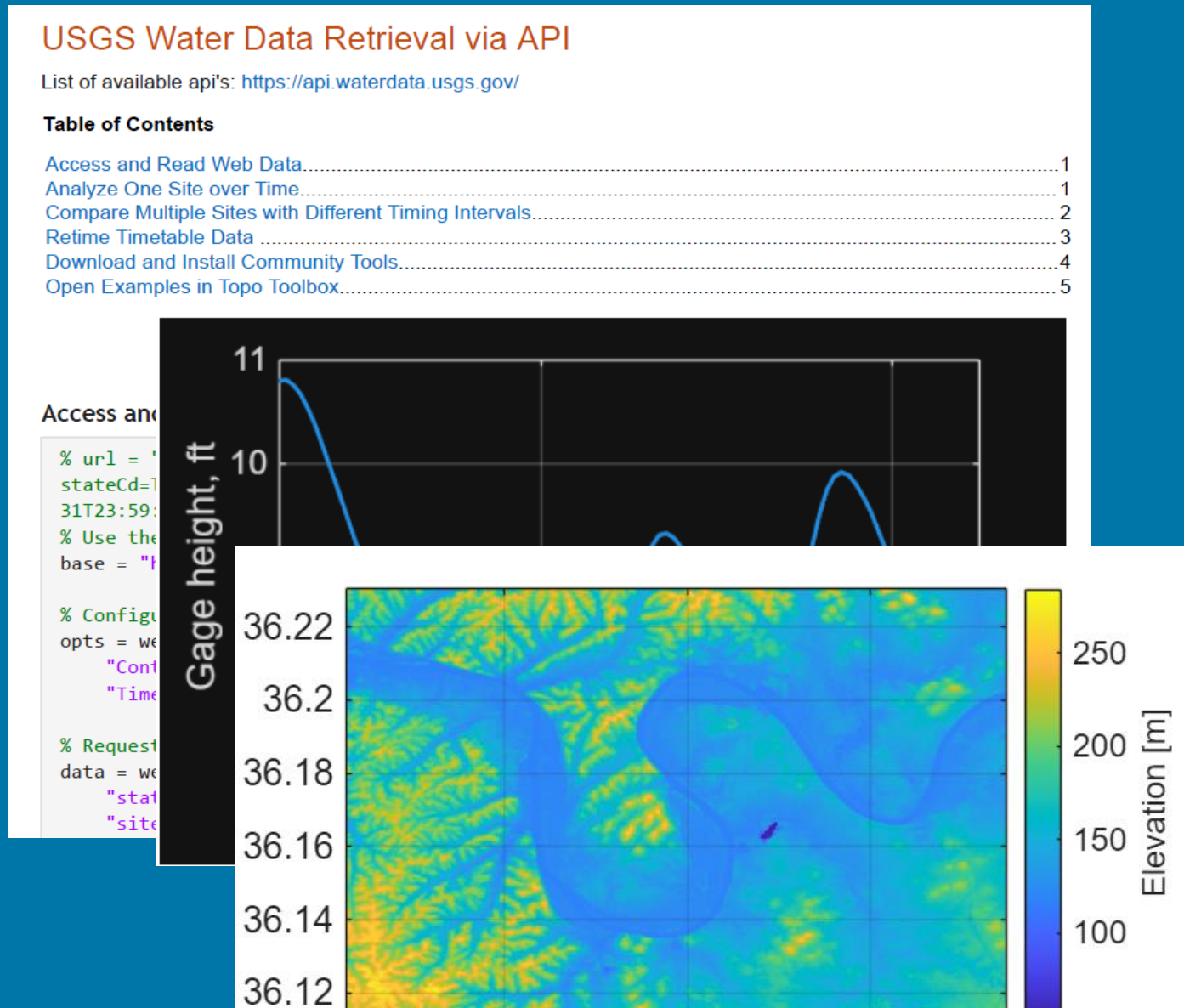
Data Processing Workflow - Example

- Easily build, run, and share your data analysis routines
  - Efficient data management
  - Informative visuals
  - Ready-to-go reports and notebooks



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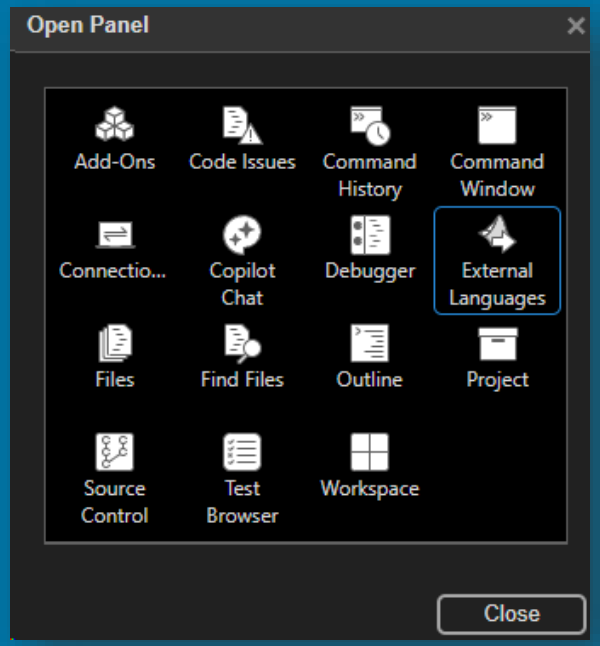
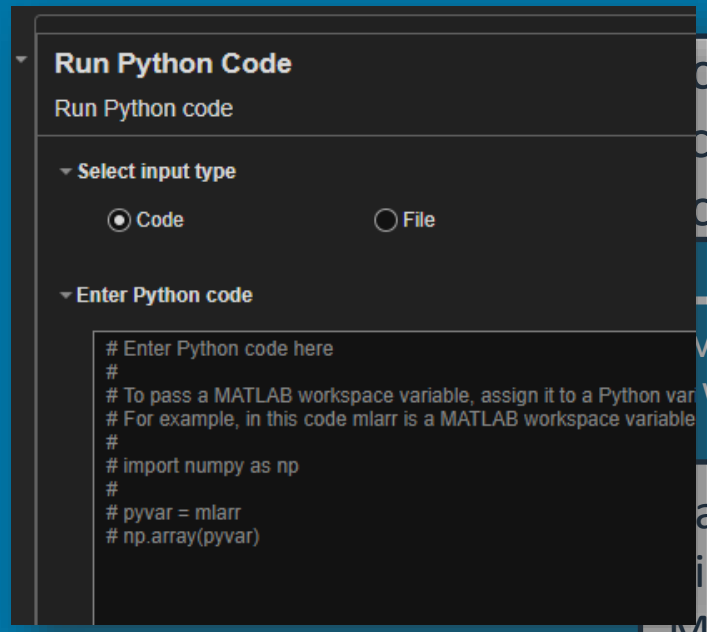
Working with Other Languages

Interoperate with Python, C, and more from MATLAB

Call Other-Language Libraries from MATLAB





cute Python
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web services

all C++ libraries
irectly from
MATLAB

Call MATLAB from Another Language

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MATLAB + R

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MATLAB R-link

Overview | Files

A COM based interface t

Original software publication

Rcall: An R interface for MATLAB

Janine Egert [a b], Clemens Kreutz [a b c]

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https://doi.org/10.1016/j.softx.2022.101276

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Directly open GitHub and File Exchange repos in MATLAB Online

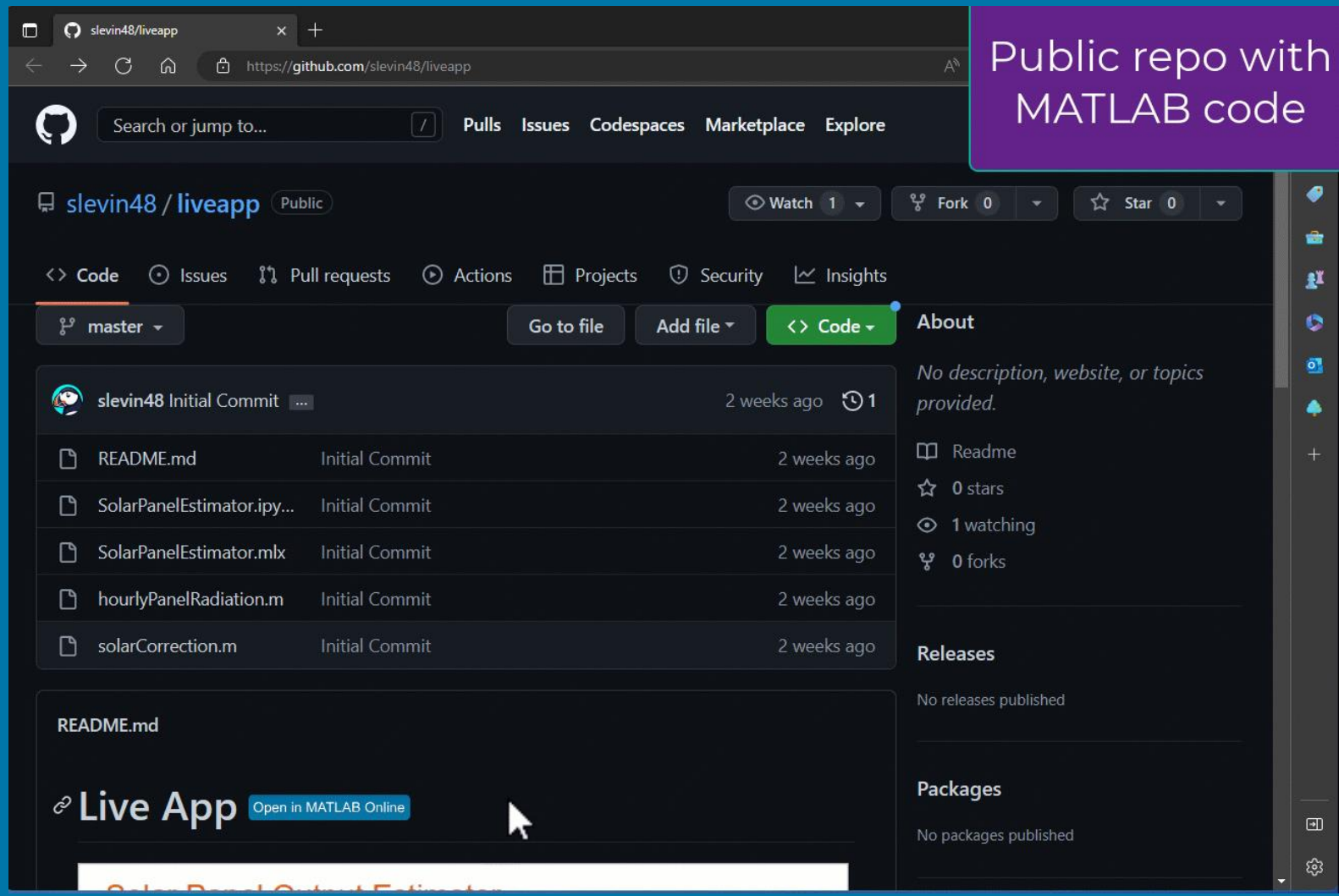
Add button to any GitHub repo

Opens in MATLAB Online

- Runs in-browser
- MATLAB Online Basic for unlicensed users
- Open in Focus Mode

Great for collaboration

- Open science
- Reproducible research
- Community toolbox building



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Use MATLAB in Jupyter Notebooks

Find on GitHub:

- https://github.com/mathworks/jupyter-matlab-proxy/

Requires MATLAB Installation

- JupyterLab and JupyerHub (Jun 2023)
- Jupyter Notebooks within VS Code (Mar 2025)

Export existing Live Scripts to Jupyter notebooks

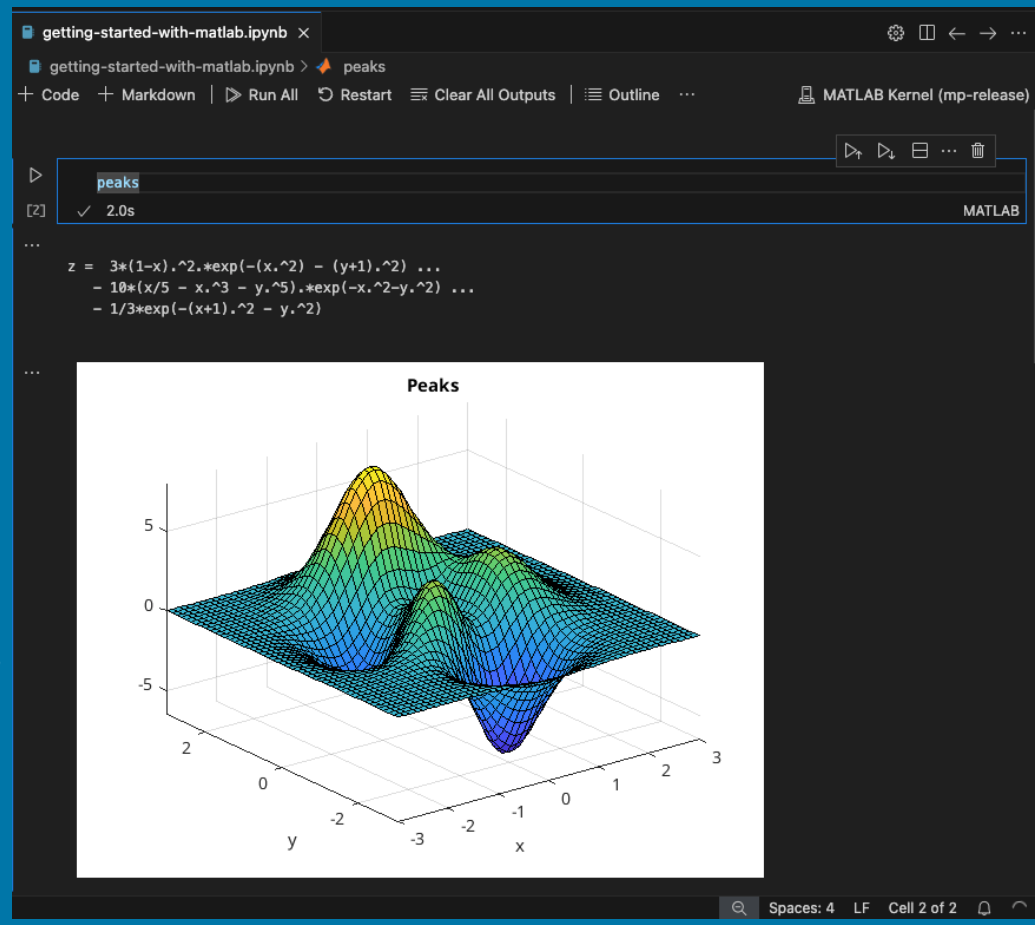

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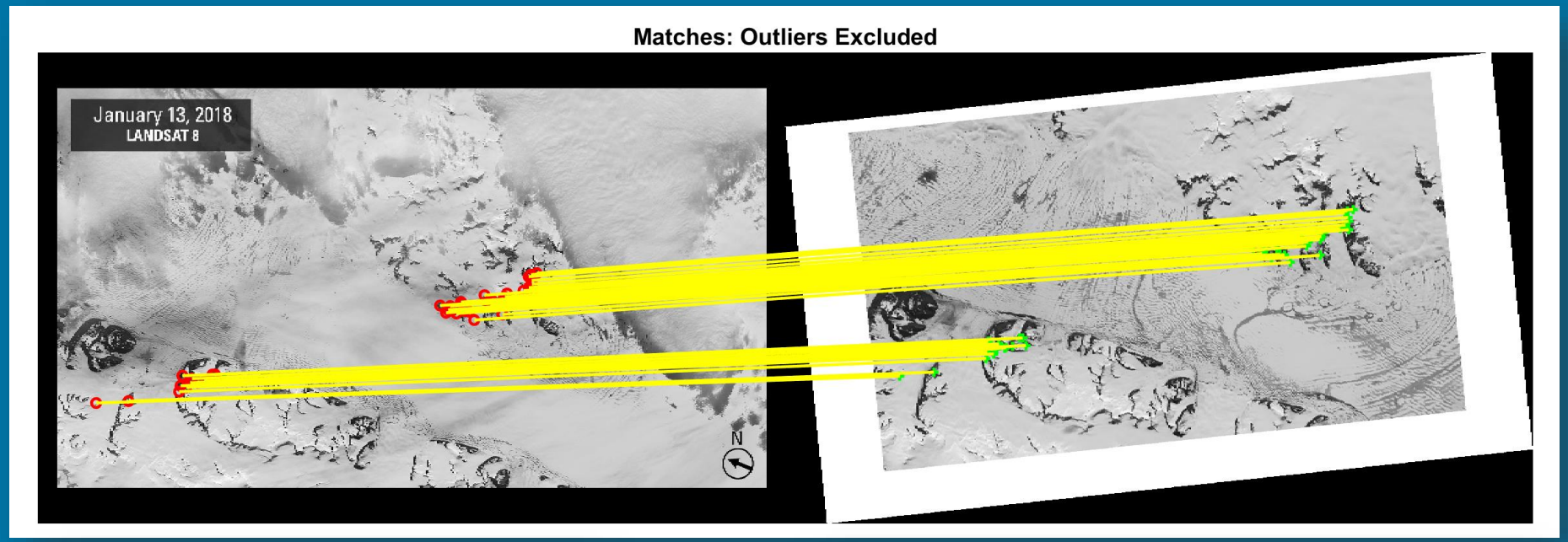
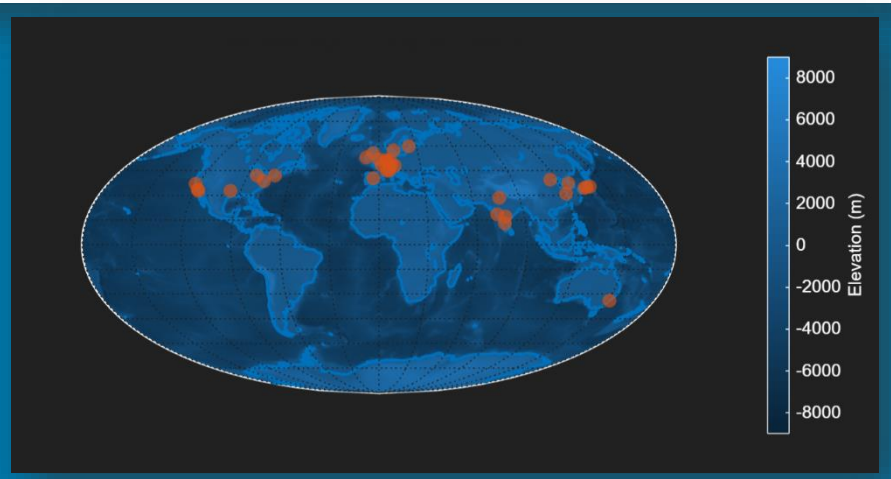
Image registration, projections, 3-D interactive maps





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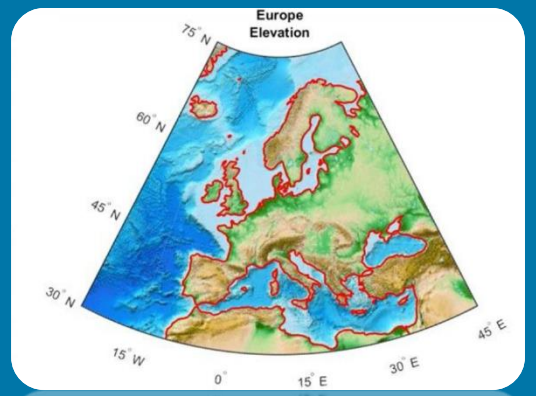
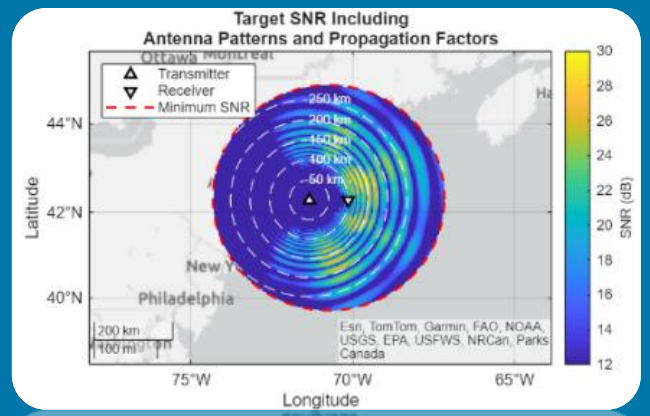


Mapping Toolbox

Radar Toolbox



Lidar Toolbox

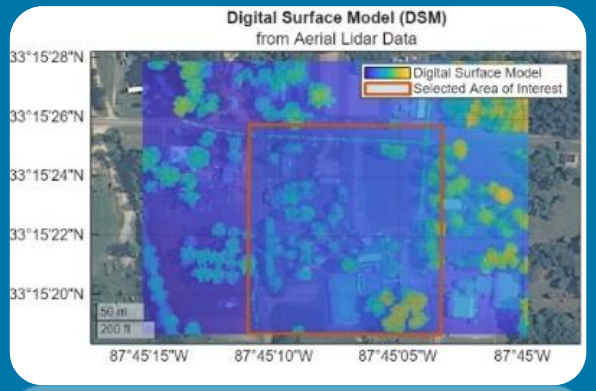
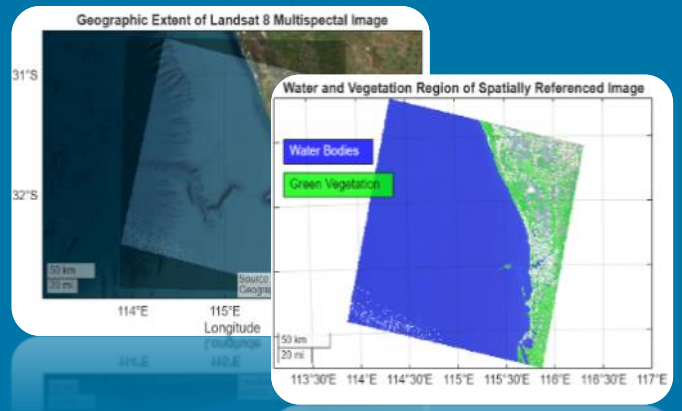


Image Processing Toolbox



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Further Learning

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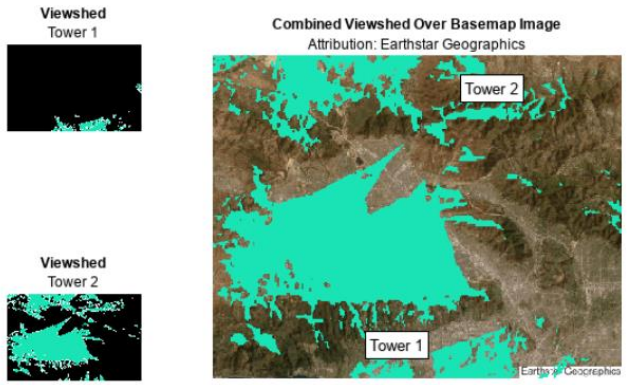
Examples and Notebooks

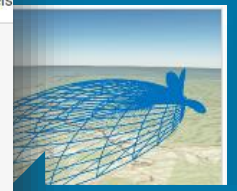
Get a jumpstart – borrow from examples

Display the basemap image and the combined viewshed on the right. Specify the z-coordinates for the labels

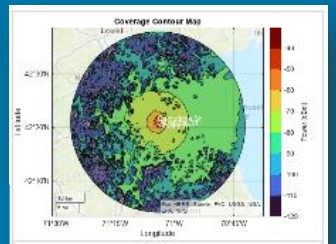
```
nexttile([2,2])
mapshow(A,RA)
hold on
mapshow(xz,yz,vis,EdgeColor="none",DisplayType="surface")
axis off

zdelta = 150;
z1 = height1 + towerHeight1 + zdelta;
z2 = height2 + towerHeight2 + zdelta;

text(x1,y1,z1,text1,Color="k",EdgeColor="k",BackgroundColor="w")
text(x2,y2,z2,text2,Color="k",EdgeColor="k",BackgroundColor="w")
title("Combined Viewshed Over Basemap Image")
subtitle("Attribution: " + attrib)
```



with Aerospace Blockset *with Hyperspectral library*

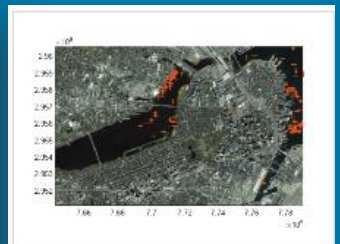


rtual Coverage
ain



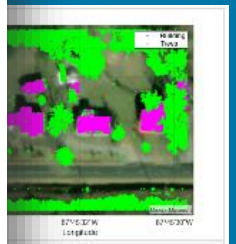
Visualize Viewsheds and Coverage Maps Using Terrain

with Antenna Toolbox



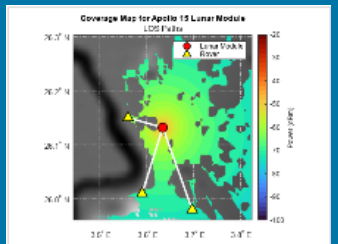
Process Large GeoTIFF File as Blocked Image

with Image Proc.Tbx



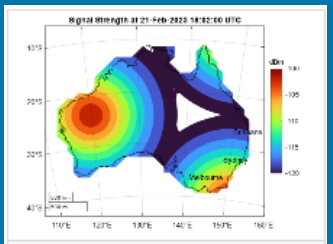
oint Cloud
tations to KML Files

with Lidar Toolbox



Visualize Coverage Maps using Ray Tracing over Lunar Terrain

with Comms Toolbox



Coverage Maps for Satellite Constellation

with SatCom Toolbox

mathworks.com/help/ *(documentation)*

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MATLAB Central Community

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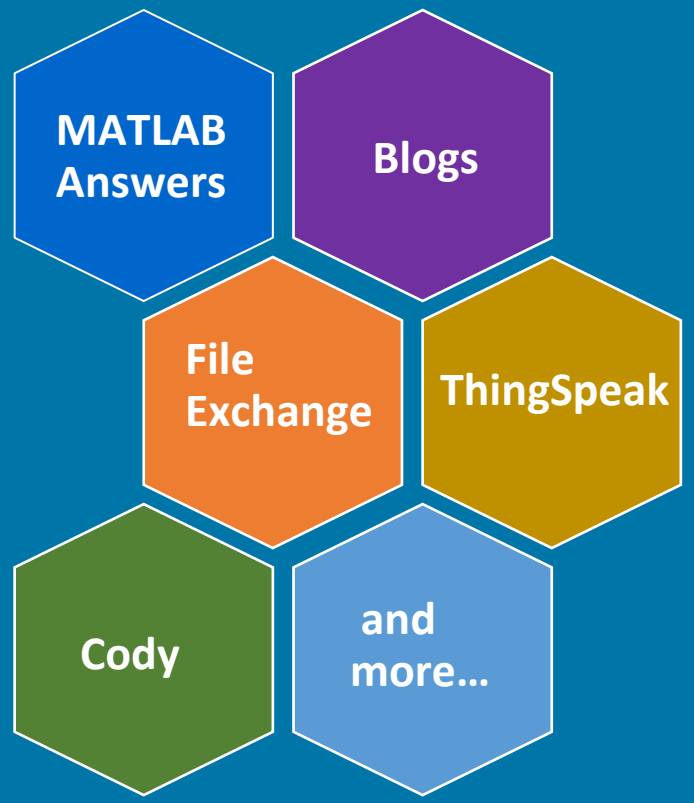
MATLAB Central

Learn | Contribute | Connect



MATLAB Answers: Q&A forum; most questions answered in <

File Exchange: Download code from a huge repository of free code including of open source community files

Cody: Sharpen programming skills while having fun

Blogs: Get the inside view from Engineers who build and support MATLAB & Simulink

ThingSpeak: Explore LoT Data

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